

The Origins of the Factor Zoo: Investors Weakly Substitute Across Stocks

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Background

- ▶ An undeniable fact: Unconstrained mean-variance $\frac{1}{\gamma}\Sigma^{-1}\mu$ implies insane portfolio weights.
Nobody does this in practice.²

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Optimal Portfolios Based on Historical Average Approach *(percent of portfolio value)*

Unconstrained							
	Germany	France	Japan	U.K.	U.S.	Canada	Australia
Currency exposure	-78.7	46.5	15.5	28.6		65.0	-5.2
Bonds	30.4	-40.7	40.4	-1.4	54.5	-95.7	-52.5
Equities	4.4	-4.4	15.5	13.3	44.0	-44.2	9.0
With constraints against shorting assets							
	Germany	France	Japan	U.K.	U.S.	Canada	Australia
Currency exposure	-160.0	115.2	18.0	23.7		77.8	-13.8
Bonds	7.6	0.0	88.8	0.0	0.0	0.0	0.0
Equities	0.0	0.0	0.0	0.0	0.0	0.0	0.0

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- ▶ Problem largely driven by inverting sample covariance matrix.
Practical solution → Shrink covariances: $\frac{1}{\gamma} (\Sigma + \lambda I)^{-1} \mu$

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This Paper

- ▶ What happens in equilibrium when we assume (more) realistic portfolio choice?³

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- ▶ Equilibrium returns

$$\mu = \underbrace{\gamma \Sigma \bar{Q}}_{\text{Risk Factors}} + \underbrace{\gamma \lambda \bar{Q}}_{\text{"Weak" Factors}}$$

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- ▶ Equilibrium returns

$$\mu = \underbrace{\gamma \Sigma \bar{Q}}_{\text{Risk Factors}} + \underbrace{\gamma \lambda \bar{Q}}_{\text{"Weak" Factors}}$$

- ▶ If λ large relative to γ we get a zoo of weak factors that have little to do with covariances!
- ▶ Intuitively: If investors don't do the crazy long-short trades implied by $\Sigma^{-1}\mu$ they don't aggressively eliminate weak factors.

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Two main insights

1. Can turn importance of risk versus constraints into a measurable object: “Substitution”, which measures how much investors reallocate between covarying stocks.

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2. Can generate factor zoo without appealing to high sharpe ratios.
Intuition: we have two degrees of freedom to control pricing of weak factors (λ) and equilibrium sharpes (γ)

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This Discussion

1. What weak substitution really means: Covariance agnostic substitution.
2. Does weak substitution cause the factor zoo?
3. A factor zoo arises if and only if there is weak substitution?
4. Suggestion: A simple argument for weak substitution from survey data.
5. (Does weak substitution necessarily imply low sharpe ratios?)

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- ▶ What low ψ really means: **Covariance agnostic substitution.**
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- ▶ Higher ridge constraint in $(\Sigma + \lambda I)^{-1}\mu$ shrinks substitution towards $1/N$
- ▶ So: covariance-agnostic substitution allows covariance-agnostic factors to persist.

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- ▶ But is $\psi = \frac{\gamma}{\lambda}$ a “deep” primitive parameter?
- ▶ λ could be large precisely because expected returns are hard to estimate.
 - ▶ Why are returns and covariances hard to estimate? Because stocks are claims to unknown cashflows far in the future... (the world is noisy, Fischer Black 1986)
- ▶ This may cause new factors to constantly appear/disappear as different investors
 - ▶ form new beliefs about the future (many factors)
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 - ▶ form new beliefs about the future (many factors)
 - ▶ Are are unwilling to take large correcting bets (low ψ)
- ▶ Weak substitution doesn't necessarily cause the factor zoo.
I'd say it is the portfolio analogue of the factor zoo.

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- ▶ Two facts from portfolio data that have received little attention in equilibrium models.⁴
 1. Investors don't hold extreme weights. → Weak substitution
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- ▶ Two facts from portfolio data that have received little attention in equilibrium models.⁴
 1. Investors don't hold extreme weights. → Weak substitution
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- ▶ The paper is about 1. But factor zoo could also arise from 2.
... as in Wurgler and Zhuravskaya (2002)
 - ▶ Idiosyncratic risk is huge. After removing 300 factors, annual $\sigma_{\text{idio}} \approx 30\%$
 - ▶ How many mispriced stocks can an investor credibly find?
 - ▶ With 50 stocks and $\bar{\alpha} = 1\%$ the sharpe is $\frac{1\%}{(30\%/\sqrt{50})} < 0.25$.
 - ▶ Weak factors persist because investors can't eliminate idiosyncratic risk.

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4. A suggestion: Weak substitution from Active shares.

- ▶ Define stock-level active share (how much investors deviate from mkt. weights) as

$$\text{Active Share}(n) \equiv \frac{\sum_i |w_i(n) - w_{mkt}(n)| A_i}{2 \times MCAP_n}$$

- ▶ Assume factor structure $\Sigma = \sigma_m^2 \beta \beta' + \sigma_{idio}^2 I$ and heterogeneous views $\mu_i = \alpha_i + \bar{\mu}$ and $\alpha_i \perp \beta$.

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- ▶ But for this to hold they must hold the investors price elasticity ζ_{own} constant, which is itself a function of ψ ...

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- ▶ But for this to hold they must hold the investors price elasticity ζ_{own} constant, which is itself a function of ψ ...
- ▶ A more reasonable claim: Weak substitution doesn't necessarily generate low sharpe ratios, but it adds an extra degree of freedom such that the model *can* generate weak factors *and* low sharpe ratios.

Conclusion

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1. The paper is rich, insightful and will be very influential. No need to overclaim!
2. Perhaps elaborate on concrete takeaway for future theory:
 - ▶ Which frictions are the most important to generate observed portfolios?
 - ▶ Leverage and individual asset short-sale constraints are not enough... (see BL 1991)

Alternative Calibration.

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Calibration. $\gamma \approx 2$, $\sigma_{idio} = 30\%$, $N = 100$, $|\alpha_i(n)| = 2\%$:

$$\text{Active Share} \approx 5.6.$$

An order of magnitude larger than the observed active shares of mutual funds.